



Full Length Article

Supportive Care

A Fructo-Oligosaccharide Prebiotic Is Well Tolerated in Adults Undergoing Allogeneic Hematopoietic Stem Cell Transplantation: A Phase I Dose-Escalation Trial



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Article history:

Received 26 April 2021

Accepted 8 July 2021

Key Words:

Hematopoietic stem cell transplantation
HCT
HSCT
BMT
Blood and Marrow Transplantation
Prebiotic
Fructo-oligosaccharides
Microbiome
GVHD
Graft-versus-host disease
Shotgun metagenomic sequencing

A B S T R A C T

Alterations of the gut microbiota after allogeneic hematopoietic cell transplantation (allo-HCT) are a key factor in the development of transplant-related complications such as graft-versus-host disease (GVHD). Interventions that preserve the gut microbiome hold promise to improve HCT-associated morbidity and mortality. Murine models demonstrate that prebiotics such as fructo-oligosaccharides (FOSs) may increase gut levels of short-chain fatty acids (SCFAs) such as butyrate and consequently induce proliferation of immunomodulatory FOXP3⁺CD4⁺ regulatory T cells (Tregs), which impact GVHD risk. We conducted a pilot phase I trial to investigate the maximum tolerated dose of FOS in patients undergoing reduced-intensity allo-HCT ($n = 15$) compared with concurrent controls ($n = 16$). We administered the FOS starting at pretransplant conditioning and continuing for a total of 21 days. We characterized the gut microbiome using shotgun metagenomic sequencing, measured stool short-chain fatty acids (SCFAs) using liquid chromatography–mass spectrometry, and determined peripheral T cell concentrations using cytometry by time-of-flight. We found that FOS was safe and well-tolerated at 10 g/d without significant adverse effects in patients undergoing allo-HCT. Community-level gut microbiota composition differed significantly on the day of transplant (day 0) between patients receiving FOS and concurrent controls; however, FOS-associated alterations of the gut microbiota were not sustained after transplant. Although the impact of FOS was fleeting, transplantation itself impacted a substantial number of taxa over time. In our small pilot trial, no significant differences were observed in gut microbial metabolic pathways, stool SCFAs, or peripheral Tregs, although Tregs trended higher in those patients who received FOS. A marker of CD4⁺ T cell activation (namely, CTLA4⁺) was significantly higher in patients receiving FOS, whereas a non-significant trend existed for FOXP3⁺CD4⁺ Treg cells, which were higher in those receiving FOS compared with controls. FOS is well tolerated at 10 g/d in patients undergoing reduced-intensity allo-HCT. Although the alterations in gut microbiota and peripheral immune cell composition in those receiving FOS are intriguing, additional studies are required to investigate the use of prebiotics in HCT recipients.

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Financial disclosure: See Acknowledgments on page 932.e10.

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INTRODUCTION

Hematopoietic cell transplantation (HCT) is a lifesaving treatment for patients with hematologic malignancies, although success remains limited by disease relapse and infectious or immunologic complications. Innovative supportive care interventions that reduce complications following HCT are essential to decrease treatment-related morbidity and

improve non-relapse mortality. The intestinal microbiome has been a major focus of research to improve outcomes following HCT and provides a uniquely modifiable target for both prevention and treatment of HCT-related complications.

The intestinal microbiome functions to maintain intestinal epithelial barrier integrity and immune homeostasis. Perturbations are associated with aberrant local and systemic immune responses that increase the risk of post-transplant complications. Decreased diversity and the presence or absence of specific intestinal organisms are associated with the development of acute graft-versus-host disease (aGVHD) and bloodstream infections (BSIs) [1,2]. The presence of *Blautia* in the gut microbiome is associated with decreased aGVHD and GVHD-related mortality [1], whereas *Enterococcus* outgrowth in the gut is associated with aGVHD [2]. Additionally, decreased relative abundance of the genera *Faecalibacterium* and *Christensenella* and the family *Barnesiellaceae* in the gut is associated with increased risk for bloodstream infections [3]. Interventions that preserve the microbiome are urgently needed to improve HCT-associated morbidity and mortality.

Diet is a key factor impacting gut microbial composition. Indigestible dietary fibers known as prebiotics serve as an important source of nutrition for the community of commensal microbes inhabiting the gastrointestinal tract. Fructo-oligosaccharides (FOSs), one of the most well-studied prebiotics, are made up of short chains of fructose (<20 in length) capped by a single glucose. FOS is tolerated up to 20 g/d or 30 g/d in healthy volunteers [4,5] and has been used in patients with inflammatory bowel disease and metabolic syndrome, among other clinical indications [4–10]. FOS and other similar prebiotic fibers are thought to support the growth of commensal organisms such as *Bifidobacterium* and to increase levels of gut microbial fermentation products known as short-chain fatty acids (SCFAs). SCFAs, such as butyrate, impact host immune regulation and maintain intestinal epithelial integrity through multiple pathways, many of which are important in preventing GVHD in HCT recipients [11–14]. For example, SCFAs are thought to induce signaling through chemokine G-protein–coupled receptors and to directly affect gene expression through histone deacetylase inhibition, resulting in the proliferation of host forkhead box P3 (FOXP3)⁺CD4⁺regulatory T cells (Tregs) [11–13] and the induction of antiinflammatory cytokines responsible for inhibiting aGVHD in murine models [14]. In a small retrospective cohort study of HCT recipients, receipt of a supplement containing 9 g/d of FOS in combination with glutamine, fiber, and oligosaccharides (GFO) was associated with significantly less GVHD and improved survival compared with those who did not receive the supplement [15]. A more recent investigation comparing HCT recipients receiving the GFO supplement to historical controls demonstrated a similar decrease in aGVHD with maintenance of gut microbial diversity with supplementation [16]. To date, no studies, to our knowledge, have comprehensively and prospectively investigated the impact of prebiotic supplementation on SCFAs, Tregs and the gut microbiome in patients undergoing HCT.

In this phase I trial, we use a prospective Bayesian optimal interval design to define the maximum tolerated dose, feasibility, and toxicity of FOS for 15 HCT patients undergoing reduced-intensity transplantation. Additionally, we compared the stool microbiome composition, peripheral immune response, and SCFA concentrations of longitudinally collected stool samples from HCT patients who received FOS to contemporaneous controls. In our study, we found that FOS was well tolerated with only minimal side effects, with intake primarily limited by transplant-related oral mucositis. We observed

differences in the gut microbiome on the day of transplant between the FOS and control groups that were not sustained at subsequent time points. We also observed an increase in activated cytotoxic T-lymphocyte-associated protein 4 (CTLA4)⁺ CD4 cells, many of which may be FOXP3⁺ Tregs, suggesting the possibility of FOS-associated immunomodulatory effects. Overall, our study provides preliminary but intriguing evidence for FOS-associated microbial and immunologic alterations that will require further study.

MATERIALS AND METHODS

Study Participants

The current study was a phase I pilot, single-arm, dose-escalation clinical trial performed at Stanford University Hospital over a period of 2 years from September 13, 2016, to December 1, 2018 (clinicaltrials.gov ID NCT02805075). A total of 15 subjects were recruited to receive FOSs throughout their transplantation. Stool samples were obtained from all 15 patients at seven time points throughout the study period (days –5, 0, +7, +14, +28, +60, and +100), and blood samples were obtained at day +28 post-transplant. An additional 16 patients undergoing the same reduced-intensity conditioning protocol without receiving FOS also participated in a longitudinal stool collection at the same seven time points; however, blood at day +28 was only available for five of these controls. All time points are referenced in the manuscript relative to the day of transplant (day 0). All participants meeting eligibility criteria were informed about the study and required to provide written informed consent under IRB #37979 prior to undergoing study-related procedures; patients not receiving the FOS consented to have their stool and blood samples collected under IRB #8903. The protocol and informed consent documents were reviewed and approved by the Stanford human subjects review board and the Stanford Cancer Institute. The study was performed in accordance with the tenets of the Declaration of Helsinki.

Patient Eligibility

Adults over 18 years of age with hematologic malignancies undergoing a reduced-intensity allogeneic HCT were eligible for enrollment. We targeted reduced-intensity allogeneic HCT recipients, reasoning that they would be more likely to tolerate oral intake compared with most myeloablative transplant recipients. No restrictions were made based on underlying malignancy or prior transplant. Patients were excluded if they had undergone gastric bypass surgery, had a history of inflammatory bowel disease or bowel obstruction, or had a known fructose intolerance or if they were enrolled in a GVHD prevention trial.

Study Design

The study was constructed to find the maximum-tolerated dose using a Bayesian optimal interval study design (Supplementary Data). The sample size was based on feasibility of accrual within a 1-year period. Successive cohorts of participants (three cohorts with five participants per cohort) were enrolled for a total of 15 participants. All 15 patients were to receive one of three possible doses (5 g/d, 10 g/d, or 15 g/d) of FOS starting on the day of hospital admission (day –5 prior to transplant) for a total of 21 days. FOS was provided for each patient in the morning as a soluble powder to be dissolved in a beverage or added to food. The dose provided to the second and third groups was dependent on the tolerability of the FOS in the preceding group. We originally defined tolerability as 80% of total doses taken. As this was an arbitrary definition, when our first patient was unable to tolerate FOS due to mucositis, we lowered the threshold of tolerability to 70%. In all subsequent subjects after this first patient, we defined tolerability as the ability to take 70% of total doses.

The FOS product used (Fibrulose F97; Cosucra Groupe, Warcoing, Belgium) is a chicory-based oligosaccharide made up of short chains of fructose (~94% of the product with <20 degrees of polymerization), joined together through $\beta(2-1)$ linkages, often ending in a unit of glucose [4]. Fibrulose F97 is a food-grade dietary supplement (Food Safety System Certification 22000), certified against microbial and heavy metal contamination. The product was handled as an investigational pharmaceutical drug according to standardized Stanford pharmacy practices to ensure patient safety.

Outcomes

Our primary objectives were to determine the maximum tolerated dose of FOS, to describe any adverse events associated with its administration, and to determine the feasibility of providing an oral supplement to patients undergoing reduced-intensity allogeneic HCT. Toxicities were graded using the Common Terminology for Adverse Events (CTCAE), version 4.0, and were tracked until 100 days after the last dose of FOS [17]. Patients also completed an adverse events survey 14 days following transplant, modeled on the National Cancer Institute's Patient-Reported Outcomes version of the CTCAE questionnaire (PRO-CTCAE) for all related gastrointestinal symptoms

(Supplementary Data) [18]. Secondary objectives included correlative analyses of intestinal microbiome and cellular effector response changes, specifically FOXP3⁺ Tregs, in patients given FOS compared with controls. Information about clinical outcomes, including grades to IV aGVHD defined according to previously established criteria [19], bloodstream infection, *Clostridioides difficile* infection, and mortality, was tracked up to 1 year following transplant.

Stool and Blood Collection

Stool samples were collected at seven time points throughout the study (days -5 ± 3 , 0 ± 3 , $+7 \pm 5$, $+14 \pm 5$, $+28 \pm 10$, $+60 \pm 15$, and $+100 \pm 30$ relative to transplant) and stored immediately at 4°C upon collection. Within 24 hours, stools were aliquoted into 2-mL cryovial tubes and stored at -80°C . Blood samples were collected in heparin collection tubes at day +28 from patients receiving FOS ($n = 14$) and from controls ($n = 5$); one FOS patient was unavailable for the day +28 blood collection. Peripheral blood mononuclear cells (PBMCs) were isolated using Ficoll (GE Healthcare Systems, Chicago, IL) density gradient centrifugation according to the manufacturer's instructions. For long-term storage, PBMCs were resuspended in FBS (Omega Scientific, Tarzana, CA) supplemented with 10% DMSO (Sigma-Aldrich, St. Louis, MO) and stored in liquid nitrogen.

Metagenomic Sequencing of Stool Samples

DNA was extracted from stool per manufacturer instructions (QIAamp DNA Stool Mini Kit; Qiagen, Hilden, Germany). An additional bead-beating step was added prior to extraction using the Mini-Beadbeater-16 (BioSpec Products, Bartlesville, OK) and 1-mm-diameter zirconia/silica beads (BioSpec Products), consisting of seven rounds of alternating 30-second bead-beating bursts followed by 30 seconds of cooling on ice. The Nextera XT DNA Library Prep Kit (Illumina, San Diego, CA) was used on DNA extractions to create sequencing libraries. When library preparations were unsuccessful using this method, we instead used the Nextera DNA Flex Library Prep Kit (Illumina) with success in all but five samples. Library concentrations were measured using Qubit Fluorometric Quantitation (Life Technologies, Carlsbad, CA). Library quality and size distributions were analyzed with the Bioanalyzer 2100 (Agilent, Santa Clara, CA). Prepared libraries were multiplexed and subjected to 100-base pair or 150-base pair paired-end sequencing on the HiSeq 4000 platform (Illumina).

SCFA Analysis From Stool Using Liquid Chromatography–Mass Spectrometry

Fecal samples were sent to the Metabolomics Core facility at the University of Pennsylvania for SCFA measurement via liquid chromatography–mass spectrometry (LC-MS). Samples were homogenized in Volatile Free Acid Mix (5 $\mu\text{L}/\text{mg}$ stool; Sigma-Aldrich) and centrifuged (13,000g for 5 minutes). The supernatant was filtered using 1.2-, 0.65-, and 0.22- μm filter plates (MilliporeSigma, Billerica, MA). The filtrate was loaded into Total Recovery Vials (Waters Corporation, Milford, MA) for analysis. SCFAs were quantified using a Waters ACQUITY UPLC System with a photodiode array detector and an autosampler (192 sample capacity). Samples were analyzed on an ACQUITY HSS T3 UPLC column (1.8 μm , 2.1×150 mm; 0.25-mL/min flow; 5- μL injection volume at 40°C ; 25-minute sample runtime). Eluent A was 100-mM sodium phosphate monobasic, pH 2.5; eluent B was methanol; the weak needle wash was 0.1% formic acid in water; the strong needle wash was 0.1% formic acid in acetonitrile; and the seal wash was 10% acetonitrile in water. The gradient was 100% eluent A for 5 minutes, with a 70% eluent B from 5 to 22 minutes, and then 100% eluent A for 3 minutes. The photodiode array was set to read absorbance at 215 nm with 4.8-nm resolution. Samples were quantified against standard curves of at least five points run in triplicate. Standard curves were run at the beginning and end of each metabolomics run. Quality control checks (blanks and standards) were run every eight samples. Results were rejected if the standards deviated by greater than $\pm 5\%$. Concentrations in the samples were calculated as the measured concentration minus the concentration of the solvent; the range of detection was at least 1 to 100 $\mu\text{mol}/\text{g}$ stool.

Immune Profiling by Mass Cytometry

Day +28 blood samples were available from 14 participants who had received FOS and five controls. Cytometry by time-of-flight (CyTOF) was performed at the Stanford Cancer Correlative Science Unit. Cryopreserved PBMCs were thawed and washed as described previously [20,21]. Cell-surface antibody master-mix or cell-staining media (CSM; $2\times$) was prepared by adding appropriate dilutions of all cell-surface antibodies into 50 μL CSM per sample. The antibody master-mix was then filtered through a 0.1-mm spin column (MilliporeSigma) to remove antibody aggregates and 50 μL was added to the sample resuspended in 50 μL of CSM. After incubation for 30 minutes at room temperature, cells were washed once with CSM. For intracellular staining, cells were fixed using the FOXP3/transcription factor staining buffer set (eBioscience, San Diego, CA) to fix for 1 hour at room temperature. After fixation, samples were washed once with CSM and once with $1\times$

permeabilization buffer (eBioscience) followed by 600g centrifugation for 5 minutes at 4°C . They were then incubated for 1 hour with intracellular markers (CD152, FOXP3, Tbet, Ki67). Intracellular antibody master mix ($2\times$) was prepared analogously to the surface antibody mix by adding appropriate dilutions of all intracellular antibodies into 50 μL permeabilization buffer per sample. Then, 50 μL of $2\times$ antibody master mix was added to the samples in 50 μL of permeabilization buffer and incubated for 1 hour at room temperature. Cells were washed once with permeabilization buffer and once with CSM. Finally, samples were resuspended in intercalation solution comprised of 1.6% paraformaldehyde in PBS and 0.5-mM iridium intercalator (Fluidigm, South San Francisco, CA) for 20 minutes at room temperature or overnight at 4°C . Before the acquisition, samples were washed once in CSM and twice in ddH₂O and filtered twice. Cells were then resuspended at 1×10^6 cells/mL in ddH₂O for acquisition on a Fluidigm Helios mass cytometer. The targeted antibodies were included in the marker panel (Supplementary Table S1) described previously [20,21]. After acquisition, data from the acquired samples were bead normalized using software based on MATLAB (MathWorks, Natick, MA) [22]. Normalized data were uploaded onto the Cytobank analysis platform to perform initial gating and population identification using previously described gating schemes [23]. Absolute cellular concentrations were calculated, $\log_{10}$ transformed, and compared between the FOS and control groups using Student's *t*-test.

Statistical Analyses

All 15 patients receiving FOS and the 16 control patients were included in the analyses. The incidence of aGVHD was compared between groups in a cumulative incidence plot using the log-rank test; overall survival was determined for all 31 patients by the Kaplan–Meier method and compared between groups using the log-rank test. Unless otherwise specified, Fisher's exact and Wilcoxon rank-sum tests were used for group comparisons of categorical and continuous variables respectively. $P < .05$ was considered statistically significant based on a two-sided analysis. All *P* values were adjusted for multiple hypothesis testing using the Benjamini–Hochberg procedure. Adjusted $P < .05$ was considered significant.

Computational Methods

Stool metagenomic sequencing data were demultiplexed by unique barcodes (bc12fastq v2.20.0.422; Illumina). Reads were trimmed using TrimGalore v0.5.0 [24], a wrapper for CutAdapt v1.18 [25], with a minimum quality score of 30 for trimming ($-q$ 30) and minimum read length of 60 ($-\text{length}$ 60). Trimmed reads were deduplicated to remove PCR and optical duplicates using seqtk rmdup v1.3-r106 with default parameters. Reads aligning to the human genome (hg19) were removed using bwa v0.7.17-r1188 [26]. To assess the microbial composition of our short-read sequencing samples, we used the Kraken v2.0.7-beta taxonomic sequence classifier with default parameters [27] and a comprehensive custom reference database containing all bacterial and archaeal genomes in GenBank assembled to “complete genome,” “chromosome,” or “scaffold” quality as of February 2019. Bracken v2.0.0 [28] was then used to re-estimate relative abundance using Bayesian statistics. Metabolic pathways were characterized using the HUMAnN2 pipeline [29]. The Shannon diversity index was calculated for each sample at the species level using the R package vegan 2.5-4 (R Foundation for Statistical Computing, Vienna, Austria). Multidimensional scaling (MDS) with Bray–Curtis distances was performed at each time point to visualize the differences in the microbial composition in the FOS and control groups using the “capscale” function from the vegan package in R. In addition, the ADONIS test was used to compare the microbial composition between the FOS and control groups at the species and genus level at baseline and each time point following transplant. Student's *t*-test was used to compare changes in the relative abundances of species at each time point (days 0, +7, +14, +28, +60, and +100) relative to baseline (day -5) between the FOS and control groups. For this analysis, only patients with fecal samples at both baseline (day -5) and later time points after transplant were included (Supplementary Figure S1). To model overall changes in the gut microbiome following transplant, mixed linear models with a second-degree polynomial for time as a fixed effect and patient ID as a random effect ($\text{taxa} \sim \text{day } 2$, $\text{random} = \sim 1 \mid \text{patient ID}$) were constructed using the “lme” function from the nlme package in R. The polynomial term in these models allowed us to model an initial decrease and then increase in the $\log_{10}$ relative abundances of taxa following transplant.

The Akaike information criterion was used to compare the fitting of the model with alternatives that did not include a polynomial ($\text{taxa} \sim \text{day}$, $\text{random} = \sim 1 \mid \text{patient ID}$), included a third- or fourth-degree polynomial ($\text{taxa} \sim \text{day } 3$ or 4 , $\text{random} = \sim 1 \mid \text{patient ID}$), and included the FOS term in the model ($\text{taxa} \sim \text{day } 2 + \text{FOS}$, $\text{random} = \sim 1 \mid \text{patient ID}$). Only taxa that were present in more than 25% of samples were included in the analysis. We chose this second-order model for simplicity and power. Although it does make strong parametric assumptions, the overall magnitude of changes seen with time have strong enough effect sizes so that significant results are robust to the assumptions. We have previously used these models to fit changes in microbial abundance over time [30]. All *P* values were adjusted for

multiple hypothesis testing using the Benjamini–Hochberg procedure. Adjusted $P < .05$ was considered significant. Similar analyses were performed to compare the metabolic pathways characterized by the HUMAnN2 pipeline between the FOS and control groups following transplant. Plots were generated in R 3.5.2 using the following packages: ggplot2 v3.1.1, genefilter v1.64.0, RColorBrewer v1.1–2, ggpubr v0.2, cowplot v0.9.4, scales v1.0.0, gtools v3.8.1, reshape2_1.4.3, dplyr_0.8.0.1, plyr_1.8.4, and ggplot2_3.1.1.

Data availability

All sequencing data sets from the current study have been deposited in the Sequence Read Archive under BioProject PRJNA746613.

RESULTS

Patient Characteristics

A total of 16 individuals undergoing HCT at Stanford were enrolled in the study and received FOS between September 2016 and October 2017 (Table 1). One study subject dropped out prior to starting the intervention; thus, a total of 15 subjects completed the study. All study subjects underwent reduced-intensity allogeneic HCT with fludarabine plus melphalan conditioning and FK506/methotrexate GVHD prophylaxis. All patients undergoing allo-HCT at Stanford received oral ciprofloxacin as antibacterial prophylaxis against Gram-negative organisms starting on the day of conditioning and continuing until engraftment. As part of the standard protocol, acyclovir and fluconazole are started on day +1 unless the patient is already on a mold-active antifungal agent. Trimethoprim–sulfamethoxazole is started for *Pneumocystis jiroveci* prophylaxis at day +42. All other medications including antibiotics were given at the discretion of the treating physician.

Table 1

Patient Characteristics Did Not Significantly Differ Between Those Receiving FOS and Controls (Wilcoxon Rank-Sum and Fisher's Exact Test, $P < .05$)

Characteristic	Group		P
	FOS (n = 15)	Control (n = 16)	
Age (yr), mean	61	65	.095
Sex, female, n (%)	7 (47)	9 (56)	.72
Race, n (%)			
White	8 (53)	11 (69)	.47
Asian	3 (20)	2 (13)	.65
Black	2 (13)	1 (6)	.60
Pacific Islander	0 (0)	0 (0)	1
Unknown	2 (13)	2 (13)	1
Ethnicity, Hispanic, n (%)	1 (7)	0 (0)	.48
Underlying malignancy, n (%)			
Acute myelogenous leukemia	5 (33)	6 (38)	1
Non-Hodgkin lymphoma	3 (20)	1 (6)	.33
Myelodysplastic syndrome	5 (33)	7 (44)	.72
Chronic myelogenous leukemia	1 (7)	1 (6)	1
Blastic plasmacytoid dendritic cell neoplasm	1 (7)	1 (6)	1
Donor type, n (%)			
Matched related	7 (47)	4 (25)	.27
Matched unrelated	7 (47)	9 (56)	.72
Mismatched unrelated	1 (7)	3 (19)	.60
Donor product, n (%)			
Peripheral blood	15 (100)	14 (88%)	.48
Bone marrow	0 (0)	2 (13%)	
Antibiotics, n (%)			
Non-carbapenem beta-lactams	12 (80)	9 (56)	.25
Carbapenems	3 (20)	5 (31)	.69
Intravenous vancomycin	9 (60)	6 (38)	.29

Stool and blood samples were collected from 15 patients receiving FOS and from 16 control patients undergoing the same reduced-intensity protocol between September 2016 and September 2018 who did not receive FOS (Table 1). Demographic and clinical characteristics of the intervention and control patients shown in Table 1 demonstrate that there were no significant differences between patients in the two groups.

FOS Was Well Tolerated

The primary goal of the study was to determine the maximum-tolerated dose of FOS in patients undergoing allogeneic HCT. The trial was constructed using the Bayesian optimal interval dose-escalation trial design with three groups of five patients receiving increasing doses of FOS depending on the tolerability of the prior group (Supplementary Data). The first group of five patients received a mean of 80% of total doses over 21 days, and four of five patients in this group met the criteria for tolerability. The next group of five patients received an increased dose of 10 g of FOS per day. Although the mean intake of this second group remained at 80% of total doses, only three of five patients met the criteria for FOS tolerability, and we were unable to advance the third and final group of five patients to the highest dose (15 g). The third group on average took 71% of total doses. We therefore found our maximum tolerated dose in this study to be 10 g of FOS per day. Intake of FOS was limited in this group by significant chemotherapy-induced nausea and mucositis, as described below.

Adverse Events Were Primarily Due to Regimen-Related Toxicity

In total, 160 adverse events in patients receiving FOS were identified, with 14 reported to be “possibly related” to FOS and the remaining thought unlikely or unrelated to the prebiotic (Table 2). A total of six patients experienced severe adverse events, including neutropenic fever ($n = 2$), disease relapse ($n = 1$), severe pain ($n = 1$), delirium ($n = 1$), and severe mucositis ($n = 1$). The most common AEs were nausea ($n = 17$) and diarrhea ($n = 16$) (Table 2). Adverse events that were reported as “possibly related” included bloating ($n = 5$), abdominal pain ($n = 4$), flatulence ($n = 4$), and constipation ($n = 1$). All of these were reported as being mild or moderate, and in only two cases did these symptoms lead to discontinuation of the FOS. Otherwise, no action was taken, or the FOS was temporarily discontinued ($n = 2$) and restarted. No adverse events were reported as being clearly attributable to FOS. Patient-reported outcomes for gastrointestinal symptoms were also assessed at day +14 in those receiving FOS and were not higher in patients receiving 10 g/d compared with those receiving 5 g/d (Table 3).

Clinical Outcomes Did Not Differ Between Groups

FOS and other carbohydrate-based prebiotics are fermented by commensal gut bacteria to produce SCFAs. SCFAs are an important source of energy for intestinal epithelial cells, essential for the maintenance of an intact intestinal mucosal barrier. They have also been demonstrated to impact immune intestinal homeostasis and increase the proliferation of Tregs, important cellular effectors for the attenuation of inflammation in the gut [31]. We hypothesized that prebiotics may have salubrious effects in decreasing aGVHD risk and therefore reducing overall mortality. We found that neither aGVHD (grades I–IV) nor overall survival at 1 year was associated with the administration of FOS compared to controls (Table 4, Supplementary Figures S2a and S2b).

Table 2
Adverse Events Reported by Cohort and Severity

Adverse Event	Cohort 1, 5 g/d (n = 5)		Cohort 2, 10 g/d (n = 5)		Cohort 3, 10 g/d (n = 5)		Total
	Grade 1-2	Grade 3-4	Grade 1-2	Grade 3-4	Grade 1-2	Grade 3-4	
Gastrointestinal							
Abdominal distension*	3	0	2	0	0	0	5
Abdominal pain*	4	0	2	0	1	0	7
Constipation*	2	0	0	0	1	0	3
Diarrhea	5	0	8	0	3	0	16
Dysphagia	0	0	0	0	2	0	2
Fecal incontinence	0	0	0	0	1	0	1
Flatulence*	2	0	2	0	0	0	4
Gastritis	0	0	1	0	0	0	1
Hiccups	1	0	0	0	0	0	1
Hyperbilirubinemia	0	0	0	0	2	0	2
Mucosal infection	0	0	0	0	1	0	1
Mucositis oral	3	0	3	1	5	0	12
Nausea	6	0	10	0	1	0	17
Oral pain	1	0	1	0	0	0	2
Stomach pain	0	0	0	0	1	0	1
Transaminitis	0	0	0	0	1	0	1
Vomiting	2	0	5	0	3	0	10
General disorders							
Fatigue	2	0	0	0	0	0	2
Fever	0	1	0	1	5	0	7
Edema in face	1	0	0	0	1	0	2
Edema in limbs	1	0	6	0	4	0	11
Pain	0	0	3	1	4	0	8
Skin							
Pruritis	0	0	0	0	1	0	1
Rash	12	0	0	0	2	0	14
Psychiatric/neurologic							
Anxiety	0	0	1	0	0	0	1
Delirium	1	0	2	0	1	1	5
Hallucinations	0	0	0	0	1	0	1
Insomnia	0	0	2	0	0	0	2
Somnolence	0	0	0	0	1	0	1
Tremor	1	0	0	0	0	0	1
Other							
Acute kidney injury	0	0	0	0	4	0	4
Allergic rhinitis	0	0	1	0	0	0	1
Atrial fibrillation	0	0	0	0	1	0	1
Epistaxis	0	0	1	0	0	0	1
Fall	0	0	1	0	1	0	2
Hyperglycemia	0	0	1	0	0	0	1
Hyperkalemia	0	0	0	0	1	0	1
Hyperthyroidism	0	0	0	0	1	0	1
International normalized ratio increased	0	0	0	0	1	0	1
Leukocytosis	0	0	1	0	0	0	1
No CTCAE term [†]	1	1	0	0	1	0	3
Syncope	0	0	0	0	1	0	1
Total	49	2	53	3	53	1	160

A higher dose of FOSs was not associated with an increase in reported adverse events.

* Reported as possibly related to FOS administration.

[†] Gout, disease relapse, and hypovitaminosis D.

Patients were monitored for infectious outcomes for 100 days post-transplant. Both *Clostridioides difficile*–associated diarrhea (CDAD) and BSIs result from an abnormal, low-diversity microbiome lacking in obligate anaerobic bacteria

[32–34]. In comparing the incidence of CDAD and BSIs in patients receiving FOS to controls, we found that CDAD occurred in only two controls (none in the FOS group), whereas six BSIs occurred in the FOS group compared with three in the

Table 3
Patient-Reported Outcomes at Day +14 After Hematopoietic Stem Cell Transplantation

Outcome	Cohort 1, 5 g/d (n = 5)		Cohort 2, 10 g/d (n = 5)		Cohort 3, 10 g/d (n = 5)		Total
	Grade 1-2	Grade 3-4	Grade 1-2	Grade 3-4	Grade 1-2	Grade 3-4	
Nausea	1	1	2	0	1	0	5
Vomiting	0	1	1	0	0	0	2
Heartburn	0	0	0	0	1	0	1
Gas	0	2	3	0	1	0	6
Bloating	0	0	3	0	1	0	4
Constipation	2	0	0	0	2	0	4
Diarrhea	3	0	2	0	0	1	6
Abdominal pain	1	0	1	0	0	0	2
Fecal incontinence	0	0	0	0	0	1	1
Total	7	4	12	0	6	2	31

A higher dose of FOSs was not associated with worse patient-reported outcomes.

*One patient did not return the survey.

control group (Table 4). Neither of these differences was statistically significant (Table 4). All nine BSIs recorded for both groups occurred with organisms thought to originate from the oral or gastrointestinal microbiome such as Enterobacteriaceae, *Streptococcus mitis*, and *Enterococcus* species.

Gut Microbiomes in Patients Receiving FOS Were Similar to Those in Control Patients

Prior studies have demonstrated greater fecal relative abundance of specific commensal bacteria, including *Bifidobacterium* spp. following administration of FOS in healthy volunteers [8]. We hypothesized that FOS would alter the gut microbiome and that patients receiving FOS would have higher Shannon diversity at day +14. Stool samples at seven time points in both FOS and control patients underwent DNA extraction, shotgun metagenomic sequencing, and taxonomic classification (Supplementary Figure S1). The number of sequenced raw reads per sample was substantially reduced following quality filtering and exclusion of human reads (median number of 28,083,873 raw reads vs. 22,173,248 median trimmed reads, 11,072,480 median deduplicated reads, and 8,129,375 filtered median non-human reads per sample) (Supplementary Table S2, Supplementary Figure S3).

We calculated Shannon diversity, a measure of microbial species richness and evenness, for each stool and compared measurements over time. In both groups, diversity declined immediately following transplant and returned to near baseline levels at day +100 (Figure 1A). To test the hypothesis that

FOS treatment was associated with shifts in the microbial community, we compared Shannon diversity among 15 subjects given FOS treatment and 16 control subjects. No significant differences in diversity were observed between patients receiving FOS and controls at any of the seven time points (Figure 1B). Additionally, MDS of stool taxonomic composition revealed that the strongest signal was not associated with receipt of FOS but instead with each individual patient ($R^2 = 38.8\%$ by ADONIS univariate test; Supplementary Figure S4a). This is consistent with many longitudinal studies of the microbiome in which the stability of the microbial community within an individual over time is a reliably strong signal [35]. The second strongest signal was with time ($R^2 = 6.2\%$), suggesting that the transplant itself was associated with a strong effect on the microbial community (Supplementary Figure S4b). The smallest signals were associated with FOS treatment ($R^2 = 1.75\%$) and FOS dose ($R^2 = 3.2\%$), suggesting that these variables were less important in explaining microbial community structure (Supplementary Figures S4c and S4d, respectively). When separated by time point, MDS revealed distinct clustering between FOS and controls at day 0 ($R^2 = 0.07$; $P = .03$, adonis test), 5 days after initiation of FOS, but not at baseline or at later time points (Figure 2). No other time points were significantly different between the FOS and control groups for either of the first two MDS axes (Supplementary Figure S5). Investigating changes in individual taxa over time relative to baseline (day -5) revealed for many taxa a pattern of decline and recovery in relative abundance (Supplementary Figure S6), similar to that of overall gut microbial diversity (Figure 2); these changes did not differ between patients receiving FOS and controls (Supplementary Figures S7 and S8). Taxa with the most significant changes following transplant included *Ruminococcus* spp., *Clostridioides* spp., and Enterobacteriaceae bacterium (Supplementary Table S3, Supplementary Figure S7).

We next examined the abundances of metabolic pathways at each time point following transplant. Unlike the taxonomic composition of the gut microbiota, the relative abundances of metabolic pathways were not different between the FOS and control groups as a whole (Supplementary Figure S9) or when compared at each time point (Supplementary Figure S10). Comparing changes in individual metabolic pathways over time relative to baseline (day -5), we found that, for the most significant metabolic pathways, relative abundance dipped only briefly after transplant and in many cases increased to levels higher than at baseline (Supplementary Figure S11).

Table 4
Transplant Outcomes Did Not Significantly Differ Between Patients Who Received FOS and Controls

Outcome	Group, n (%)		P
	FOS (n = 15)	Controls (n = 16)	
Acute GVHD (grades I-IV)	9 (60)	9 (56)	1
Bloodstream infections	6 (40)	3 (19)	.25
<i>Clostridioides difficile</i> infection	0 (0)	2 (13)	.48
Overall mortality			
6-month	2 (13)	2 (13)	1
12-month	3 (20)	6 (38)	.42

FOS was not associated with significant differences in acute GVHD, bloodstream infection, *Clostridioides difficile* infection, or 6-month and 12-month mortality compared with control patients (Fisher's exact test, $P < .05$).

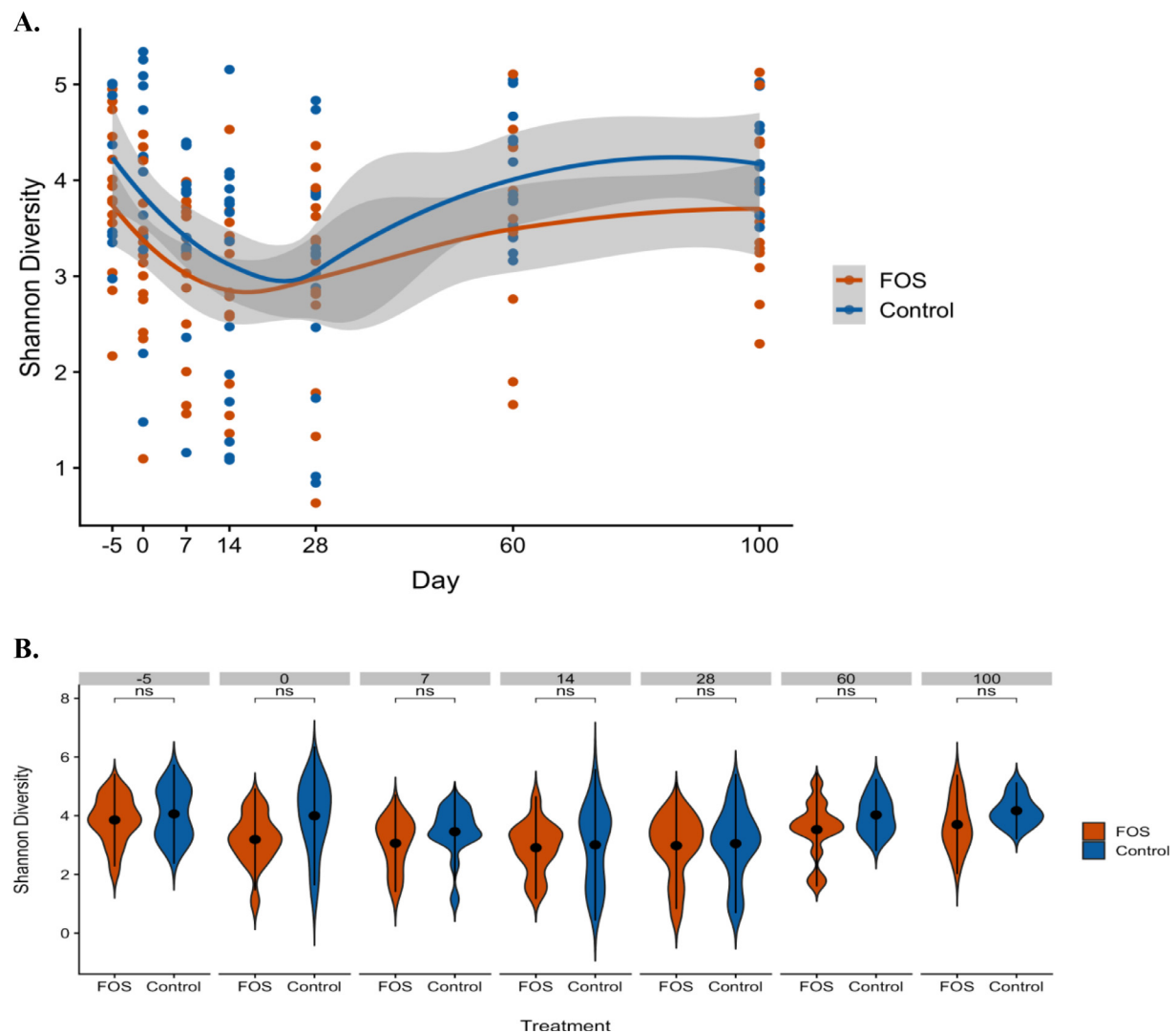


Figure 1. FOS treatment was not associated with changes in alpha diversity relative to controls. Shannon diversity, a measure of species richness and evenness, was calculated for each microbiome sample. (A) Shannon diversity declined after transplant but was restored by day 100 post-transplant. Regression lines were fit using LOESS smoothing, with gray bands representing 95% confidence intervals. (B) Shannon diversity was not significantly different between FOS-treated patients and controls at any time point (Wilcoxon rank-sum test, false discovery rate-adjusted $q < .05$).

SCFAs Did Not Differ Among Patients Based on FOS Intake

We hypothesized that those patients receiving FOS would have increased levels of SCFAs in their stool potentially due to enrichment in organisms that ferment FOS to produce SCFAs such as butyrate. Unsurprisingly, given the lack of sustained enrichment of microbial taxa over time, day +14 stools analyzed using LC-MS did not demonstrate significant differences in levels of butyrate, propionate, or acetate (Supplementary Figure S12). CyTOF of peripheral blood demonstrated higher levels of CD4⁺ T cell activation and a trend toward higher FOXP3⁺CD4⁺ Treg concentrations in patients receiving FOS compared with controls.

Microbially produced SCFAs impact host immune responses in part through induction of intestinal Tregs. SCFAs such as butyrate induce the colonic differentiation of CD4⁺FOXP3⁺ Tregs, crucial to host intestinal immune homeostasis [11,12]. We therefore hypothesized that the administration of FOS would impact peripheral Treg concentrations at day +28 when compared with controls. To determine whether FOS intake was associated with higher

concentrations of peripheral Tregs after transplantation, blood samples were compared between 14 patients who received FOS and five controls to determine whether FOS was associated with alterations in peripheral immune reconstitution at day +28 following hematopoietic stem cell transplantation compared with controls. Whole-blood-derived PBMCs were subjected to a previously validated, comprehensive CyTOF workflow developed for identifying immune signatures for GVHD in peripheral blood following HCT [20,21]. Complete results comparing median log₁₀-transformed absolute cell concentrations between the FOS group ($n = 13$) and the control group ($n = 5$) are shown in Supplementary Figure S13 (cell counts from one participant who had received FOS were discarded due to an inadequate number of cells). Although median concentrations of CD4⁺FOXP3⁺ Tregs differed to a great extent between FOS-treated patients and controls (2.1 vs. 0.6 cells/ μ L), this difference was not statistically significant after adjusting for multiple hypothesis testing (unadjusted $P = .0129$, adjusted $P = 0.145$) (Figure 3). Of note, CTLA4, a

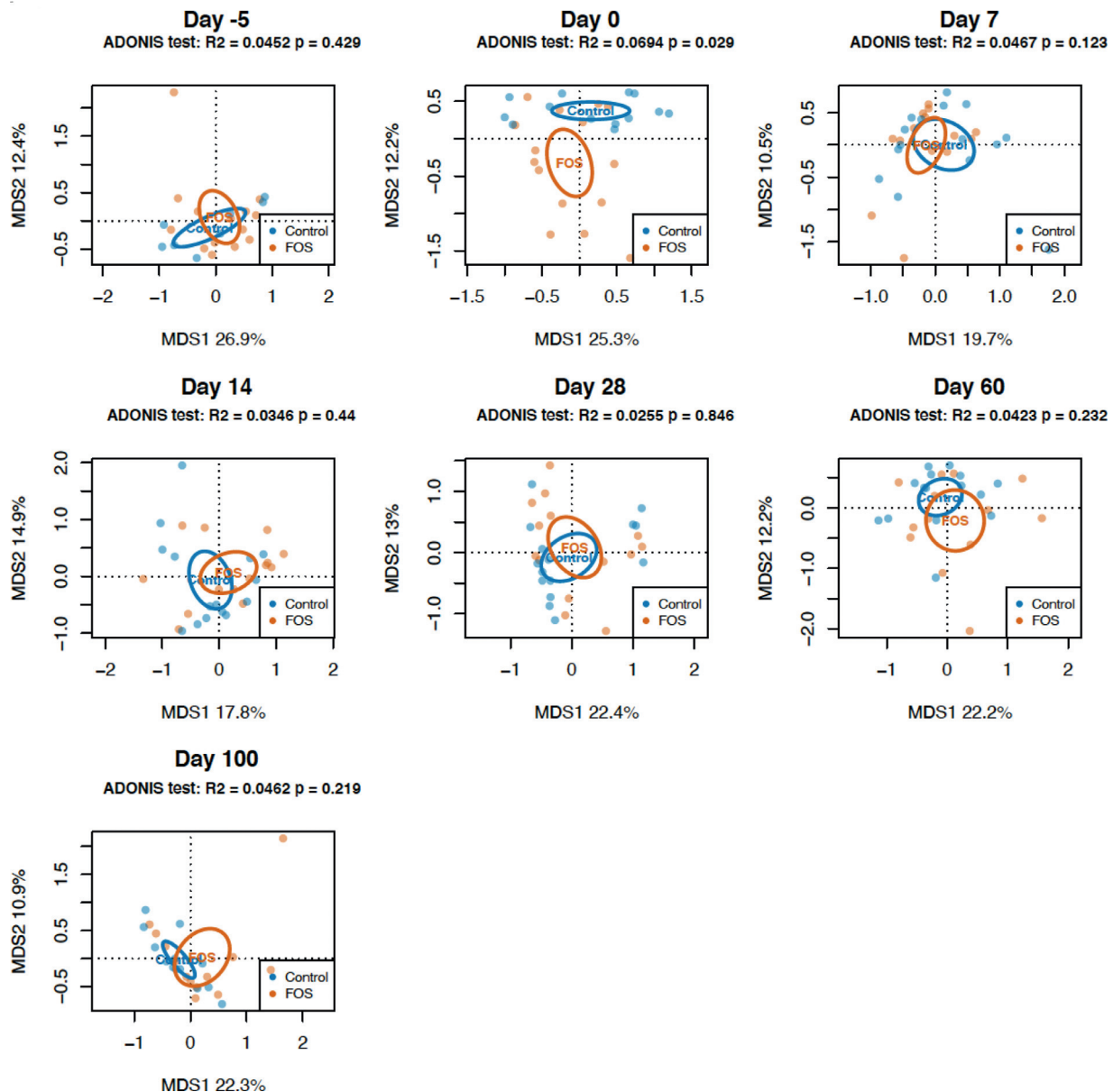


Figure 2. Gut microbiota composition was significantly different between the FOS and control groups only at day 0, on the day of transplantation. MDS of gut microbiota taxonomic composition at the species level was performed using Bray–Curtis distances at each time point relative to transplantation. The adonis test was used to compare the gut microbiota composition between the FOS and control groups at each time point. At day 0, the day of transplant, the FOS and control groups were significantly associated with the second axis of the ordination plot, which explains 12.2% of the variation at this time point (simple linear regression, $R^2 = 0.31$, adjusted $P = .03$).

marker of CD4⁺ T cell activation often associated with Tregs, was found to be significantly different between the FOS and control groups (adjusted $P = .013$). We used Student's t -test to compare those patients who received 5 g ($n = 4$) versus 10 g ($n = 13$) of FOS and did not observe any differences in CTLA4⁺CD4⁺ or FOXP3⁺CD4⁺ T cell concentrations (data not shown). Other immune responses following transplantation were not statistically significant after adjustment for multiple comparisons, including markers of naïve and memory T cells (measured by CD45RA and CCR7) (Supplementary Figure S13). Nevertheless, these data suggest an association between FOS and Treg biology that supports our hypothesis, despite the lack of statistical power in our study to demonstrate a clinical impact on aGVHD.

DISCUSSION

The microbiome has emerged as a promising potential contributor to allo-HCT patient outcomes. However, to date, few prospective interventional trials have investigated the effects of microbiome-modifying therapies (e.g., diet, antibiotics, prebiotics, probiotics) for their potential impact on outcomes such as aGVHD incidence and overall survival. In this study, we performed a prospective trial of prebiotics in HCT recipients with concurrent controls. We found that FOS was well tolerated at a maximum tolerated dose of 10 g/d, limited only by chemotherapy-induced nausea and mucositis. Patients reported mild abdominal distension and flatulence, and in only two patients were these effects dose limiting. No significant differences in adverse outcomes, including GVHD, infection, or mortality, were observed in association with FOS administration,

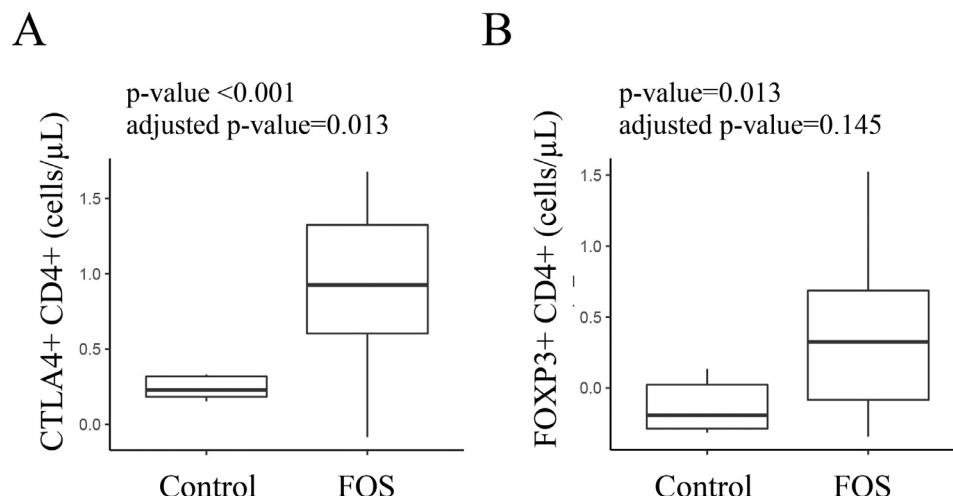


Figure 3. Patients receiving FOSs during hematopoietic stem cell transplantation had higher levels of peripheral CTLA4⁺ and FOXP3⁺CD4⁺ T cells compared with controls at day +28. CyTOF was used to determine peripheral concentrations of specific T cells at day +28 following transplantation. Absolute concentrations were log₁₀ transformed and compared between the FOS and control groups using Student's *t*-test. CTLA4⁺CD4⁺ T cells were significantly higher in patients receiving FOS after adjusting for multiple comparisons using Benjamini–Hochberg correction. FOXP3⁺CD4⁺ T cells were also higher in those receiving FOS compared with controls, although this was not statistically significant after FDR correction.

although our study was underpowered to detect small differences.

In our pilot, we found that FOS was associated with community-level taxonomic differences on the day of transplant (day 0). These differences between groups did not endure beyond transplantation, perhaps as a result of broad-spectrum antibiotic use that occurs with febrile neutropenia treatment. Following day 0, a profound change in the microbial community was observed, consistent with prior literature [36]; as many as 20% of the taxa were altered while often recovering to pre-transplant levels, albeit slowly over 100 days (Supplementary Figure S6). These data suggest that any signal observed as a result of FOS was temporary and likely overwhelmed by other factors, including antibiotics, and dietary intake following transplantation.

In addition to a modest short-term microbial signal associated with FOS intake, there was a higher concentration of CTLA4⁺CD4⁺ T cells and a non-significant trend of higher peripheral FOXP3⁺CD4⁺ Tregs in those who received FOS compared with controls, which had been suggested previously by prior murine models [37]. Given that these CTLA4⁺ cells were CD4⁺ T cells, it is likely that a significant proportion are activated Tregs. Although larger randomized studies are needed to validate this finding, the possibility that activated Tregs may be increased by a dietary supplement is quite intriguing. If it is more definitively demonstrated to have an impact on CTLA4⁺ Treg concentrations, prebiotic supplementation may play an important role in efforts to prevent acute GVHD through boosting Treg concentrations following HCT, a process that has proven very challenging when performed using pretransplant ex vivo manipulation of the stem cell graft [38].

The limitations of our study are multiple, including that we performed our investigation at a single center. As our study was not powered to detect differences among groups resulting from FOS intake, our ability to observe differences in taxonomic composition and clinical outcomes was greatly limited due to sample size. Despite having a limited sample size, we performed extensive correlational analyses based on expected alterations in the gut microbiome, SCFAs, and Treg concentrations in order to inform future, larger studies. In addition, FOS was not taken consistently throughout the post-transplant period due to

severe mucositis. Intermittent or truncated prebiotic intake may have impacted our ability to detect changes in microbiome composition or clinical outcomes compared with prior studies demonstrating significant prebiotic efficacy, in which adherence was more consistent and FOS was administered for a longer duration both before and after transplantation (21 vs. 35 days) [15,16]. Antibiotics administered before transplant may have also eliminated many of the bacteria capable of using FOS for nutrition. Indeed, Yoshifuji and colleagues [16] demonstrated that patients with a higher baseline microbiome diversity appeared more likely to benefit from prebiotic supplementation, potentially due to the greater presence of prebiotic fermenting bacteria. In a prior study in which FOS intake was associated with decreased rates of aGVHD [15], patients received not only FOS but also a *Lactobacillus* probiotic as part of standard of care. The use of a “synbiotic,” combining a prebiotic with a probiotic, may be more efficacious in improving clinical outcomes and positively impacting the microbiome compared with using a prebiotic alone, as demonstrated in other gastrointestinal disorders such as inflammatory bowel disease [7]. In two prior published studies investigating FOS use in HCT recipients [15,16], supplementation of dietary fibers with resistant starches in combination with oligosaccharides may have been more effective at maintaining gut microbial diversity and improving clinical outcomes. Longer duration of prebiotic use both before and after HCT along with multifiber supplementation may therefore demonstrate greater prebiotic efficacy and more significant, durable maintenance of the baseline microbiota throughout transplantation.

Although our findings demonstrated a limited community-level microbial signal on the day of transplant associated with FOS intake, the pre- and peritransplant microbiome has exhibited significant associations with favorable outcomes following HCT [34,39]. To maintain a greater proportion of the baseline microbiota, further investigation of preemptive microbiota-targeting therapies given for a longer duration prior to transplantation or conditioning in larger randomized studies is worthwhile. Additionally, therapies targeting a greater number of microbes may require more than one prebiotic at a time with or without a probiotic/synbiotic, capable of maintaining microbiome diversity and essential commensal bacteria

throughout transplant. It may be beneficial to determine whether organisms capable of fermenting specific prebiotics are present in the pre-transplant gut microbiome before patients undergo allo-HCT. For patients lacking the necessary commensals, personalized synbiotic cocktails composed of SCFA-producing bacteria maintained via prebiotic administration may be required to supplement baseline gut flora prior to transplant. Personalization of microbiome-altering therapies and supplementation of the gut microbiome prior to transplant—in combination with judicious antibiotic use to maintain microbial diversity—may offer a unique and promising method of mitigating complications from stem cell transplantation and improving survival outcomes.

ACKNOWLEDGMENTS

We thank the members of the Bhatt Lab for providing feedback on the study design, bioinformatics pipeline, and manuscript revisions. We especially thank the patients and nurses on the Blood and Marrow Transplantation service for their enthusiastic participation in this project. This work was supported in part by an Amy Strelzer Manasevit Award from the National Marrow Donor Program; by National Center for Advancing Translational Science, Clinical and Translational Science Awards (KL2TR001083 and UL1TR001085); and by a Society of Blood and Marrow Transplantation New Investigator Award (T.M.A.). F.B.T. was supported in part by a National Science Foundation Graduate Research Fellowship. A.S.B. was supported by a National Cancer Institute Award (KO8 CA184420), the Amy Strelzer Manasevit Award, a Damon Runyon Clinical Investigator Award, and a V Foundation Scholar Award. The funding source for the cytometry by time-of-flight (CyTOF) experiments was the Parker Institute for Cancer Immunotherapy. The CyTOF experiments were reviewed by Crystal Mackall, MD, and the Stanford Cancer Correlative Science Unit leadership and approved for processing.

Financial disclosure: Cosucra (Warcoing, Belgium) provided the fructo-oligosaccharides free of charge but were not involved in the design, execution, or analysis of the study. Fructo-oligosaccharides have received the Generally Recognized as Safe designation and therefore did not require a US Food and Drug Investigational New Drug Application.

Conflict of interest statement: There are no conflicts of interest to report.

Authorship statement: A.A.F., A.R.R., and A.S.B. contributed equally to this work.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jctc.2021.07.009.

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